

Dual Robot Avatar: Real-time Multispace Experience using Telepresence Robots and Walk Sensation Feedback including Viewpoint Sharing for Immersive Virtual Tours

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ABSTRACT

Traveling to different places simultaneously is a dream for several people, but it is difficult to realize this aspiration because of our physical space limits. On one hand, virtual reality technologies can help alleviate such limits. According to the best of the authors' knowledge, there is no study attempt to operate multiple telepresence robots in remote places simultaneously, with presenting walk sensation feedback to the operator for an immersive multispace experience. In this study, we used autonomous mobile robots; a dog and wheel type one, where their movements' direction can be controlled by an operator (Fig. 1). The operator can alternatively choose/re-choose the space (or robot) to attend and can move the viewpoint using a head-mounted display (HMD) controller. A live video image with 4 K resolution is transmitted to the HMD via web real-time communication (WebRTC) network from a 360° camera placed to the top of each robot. The operator perceives viewpoint movement feedback as a visual cue and vestibular feeling via waist motion and proprioception on the legs. Our system also allows viewpoint sharing in which fifty users can enjoy omnidirectional viewing of the remote environments through the HMD without walk-like sensation feedback.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; *Haptic devices*.

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SIGGRAPH '22 Emerging Technologies, August 07-11, 2022, Vancouver, BC, Canada

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ACM ISBN 978-1-4503-9363-8/22/08.

<https://doi.org/10.1145/3532721.3535570>

KEYWORDS

Multi-space walking, avatar robots, virtual tours, walking sensation, teleoperation, virtual reality.

ACM Reference Format:

Yusuke Kikuchi, Yukiya Ojima, Ryoto Kato, Minori Unno, Vibol Yem, Yukie Nagai, and Yasushi Ikei. 2022. Dual Robot Avatar: Real-time Multispace Experience using Telepresence Robots and Walk Sensation Feedback including Viewpoint Sharing for Immersive Virtual Tours. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Emerging Technologies (SIGGRAPH '22 Emerging Technologies)*, August 07-11, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532721.3535570>

1 INTRODUCTION

Travel experience is important for everyone regardless of age or sex. Due to the current issue of a pandemic disease COVID-19, various virtual tour services have been provided which travelers can view the live video of sites that is broadcasted by a tour agency. However, the lack of activity and vestibular or proprioceptive sensation feedback reduce the sensation of immersive experience. Various motion simulators have been developed as described in [Bruck et al. 2021]. However, most of these are for experience in computer graphics space rather than in the real-world environment, and the amount of stimulator motion relative to the telepresence robot remains an issue. Telepresence or Telexistence is a concept of an immersive experience during the control of a remote robot by using the somatic sensation to the operator [Suomalainen et al. 2022] [Tachi 2016]. In the previous study, we imparted the acceleration sensation of a remote mobile vehicle to a local user by a wheel-chair type simulator [Yem et al. 2019]. However, the previous studies attempted to provide an experience in one space, which is not sufficient for rich information experience in various spaces.

In a recent study, we overcame the issue of space limitations by enabling users to experience various remote spaces in real-time using multiple semi-autonomous telepresence robots (Fig. 1). The

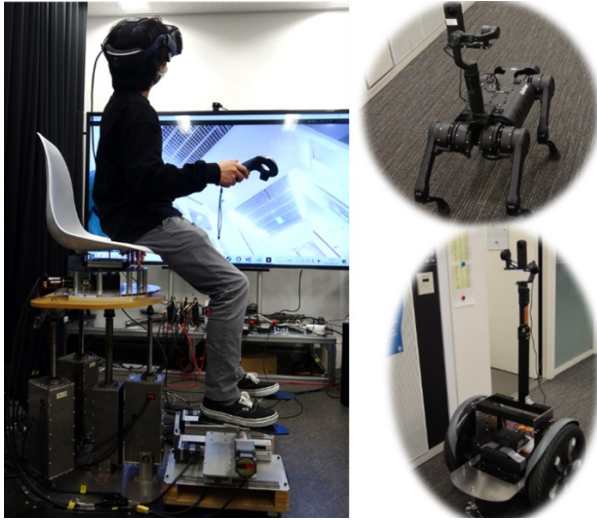


Figure 1: Multispace experiencing in real-time with walk sensation feedback

operator can choose/re-choose the space to attend during the experience using a touch controller. The operator perceives vestibular sensation by motion at the waist and proprioception at the legs to increase the immersion of walking in a remote environment during moving a viewpoint. Our recent system, Dual Robot Avatar (an avatar composed of two robots) can share the viewpoints of two telepresence robots to fifty users, which is suitable for virtual tours in two different spaces at the same time.

2 SYSTEM

2.1 Walking sensation feedback:

The detail of a walk display is described in [Unno et al. 2021]. A user sitting on the motion seat can feel the vestibular sensation by the waist motions with four Degrees of Freedom (DoF) (e.g., lift, yaw, pitch, and roll). Two pedals, each with three DoF, move the heels up/down, forward/backward, and left/right. These motions were not sufficient for generating various kinds of bodily sensations, but they are the basic motions to generate walking activities.

2.2 Telepresence robots:

In our recent system, we used dog (A1, Unitree Robotics) and wheel (RMP 200, Segway) type robots that are controlled using a compact AI computer (Jetson Xavier NX with Jetpack Version 4. 6, NVIDIA). To make our robot compact, the same computer executes data transmission and live video broadcasting. A 360° camera with 4 K resolution (THETA Z1, Ricoh) and a stabilizer were attached to each robot. Each robot contains a depth camera (RealSense D435i, Intel) that can be used for obstacle detection.

2.3 Viewpoint sharing and system configuration:

We used a cloud server with WebRTC Selective Forwarding Unit (SFU), for data and media transmission between the operator and

robots. This service allows a hundred one-way channels (user connections), therefore fifty users can join in two remote spaces at the same time. In addition, each space has the same viewpoint, moved by a dedicated operator, to approximately fifty participants. Due to a limited number of walk displays, vestibular and proprioceptive sensation feedback are provided to a single user who operates the robots. Figure 2 shows our system configuration. The operator uses buttons on a touch controller to choose/re-choose robot and moves the robot using the joystick.

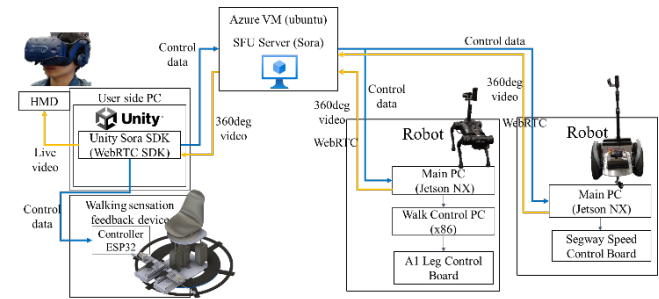


Figure 2: System configuration

3 DEMONSTRATION

An attendee sits on a seat-type walk display and controls two telepresence robots alternately using a touch controller. The robots run inside our exhibition booth. Although the robots are autonomous, embedded with obstacle detection sensors, two of exhibitors will stay nearby each robot and hold an emergency button to avoid safety hazards. Moreover, other attendees can view omnidirectionally the robot environments with the same viewpoint of the operator via an HMD without walk sensation feedback when they prefer only visual cues.

ACKNOWLEDGMENTS

This work was partially supported by Local-5G Project at TMU, MIC/SCOPE #191603003 and JSPS KAKENHI Grant Number 18H04118.

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